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Hola!
This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

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$$ \textsf{\huge Water Viscosity, μ_w} $$

1. Introduction

The viscosity of formation water, μ_w, is a measure of the resistance of water to flow. In reservoir engineering, water viscosity is commonly expressed in centipoise (cP) and is an important property for determining water mobility through porous media.

Water viscosity is primarily affected by temperature, pressure, salinity, and dissolved-gas content. In general, its variation with pressure is considerably smaller than that observed for oil viscosity.

At a constant reservoir temperature, μ_w, generally decreases as reservoir pressure decreases. This behavior occurs both above and below the bubble-point pressure, p_b. However, a small change in the slope of the μ_w -versus-pressure relationship may occur at p_b because of changes in the amount of gas dissolved in the formation water.

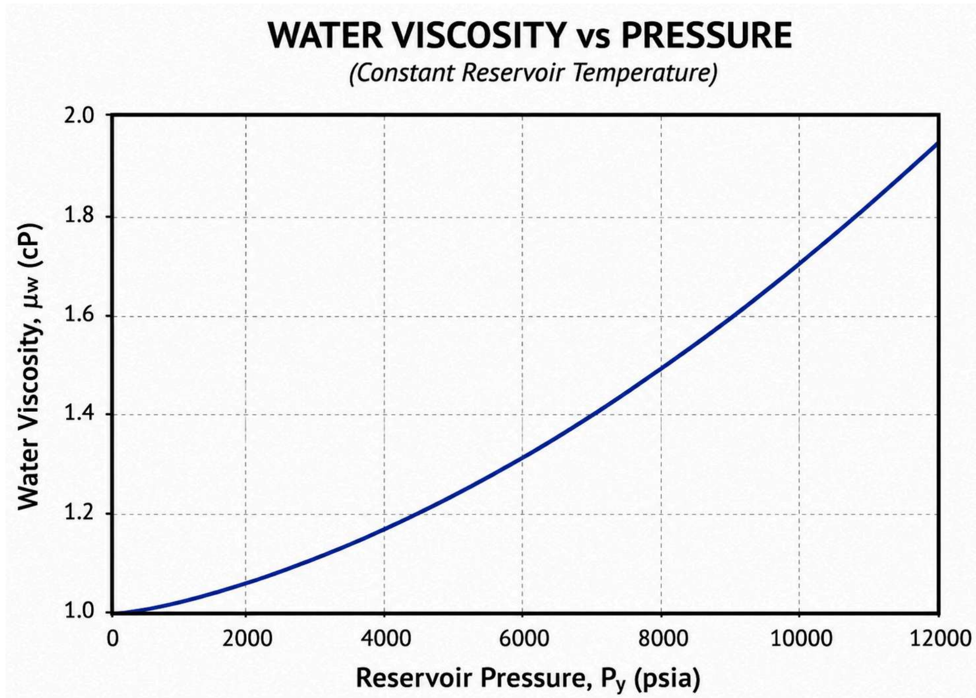
- For $p>p_b$, the water remains undersaturated with respect to gas, and a decrease in pressure causes a slight decrease in water viscosity. Because water is only slightly compressible, the pressure effect in this region is relatively small.
- At $p=p_b$, the water becomes saturated with dissolved gas. The transition through the bubble point may produce a subtle change in the pressure dependence of μ_w.
- For $p<p_b$, gas begins to come out of solution as pressure decreases. Nevertheless, because the solubility of natural gas in water is relatively low, this effect on water viscosity is generally small. Thus, μ_w, continues to decrease gradually as pressure decreases.

In practice, temperature (T) and salinity (S) often have a more significant influence on water viscosity than pressure. Increasing temperature substantially decreases μ_w, whereas increasing salinity generally increases the viscosity of formation water.

$$ \mu_w =f(T,S,p,...) $$

Unlike oil viscosity, μ_o, water viscosity does not exhibit a pronounced minimum around the bubble-point pressure, p_b. This difference is primarily due to the much lower solubility of natural gas in water. Because only a small amount of gas is dissolved in the aqueous phase, gas liberation below p_b produces only a minor change in water viscosity. Consequently, the variation of μ_w with pressure is generally smooth and gradual, with only a slight change in slope near p_b. Over a large pressure interval, however, the cumulative effect can become significant. For example, μ_w may decrease to approximately half of its high-pressure value when pressure decreases from about 12,000 to 1,000 psia, depending on temperature, salinity, and fluid composition.

Overall, formation-water viscosity is controlled mainly by temperature, salinity, and pressure, while the effect of dissolved gas is usually secondary. The following figure shows the typical variation of the water viscosity as a function of pressure.



There are several numerical correlations available to compute the oil formation volume factor, B_o, some of the most widely used include:

- Van Wingen
- Matthews
- McCain

2. Numerical solution

$ \text{viscosity at 1 atm} $

$$ \mu_{w1} = AT^{B} $$

$$ A=109.574-8.40564S+0.313314S^{2}+8.72213 \times 10^{-3} S^{3} $$

$$ B=-1.12166+2.63951 \times 10^{-2} S-6.79461\times 10^{-4} S^{2}-5.47119\times 10^{-5} S^{3}+1.55586\times 10^{-6} S^{4} $$

$ \text{water viscosity at p $\geq$ 1 atm} $

$$ \mu_{w} = (0.9994+4.0295\times 10^{-5}p+3.1062\times 10^{-9} p^{2}) \mu_{w1} $$

2.1 Import libaries

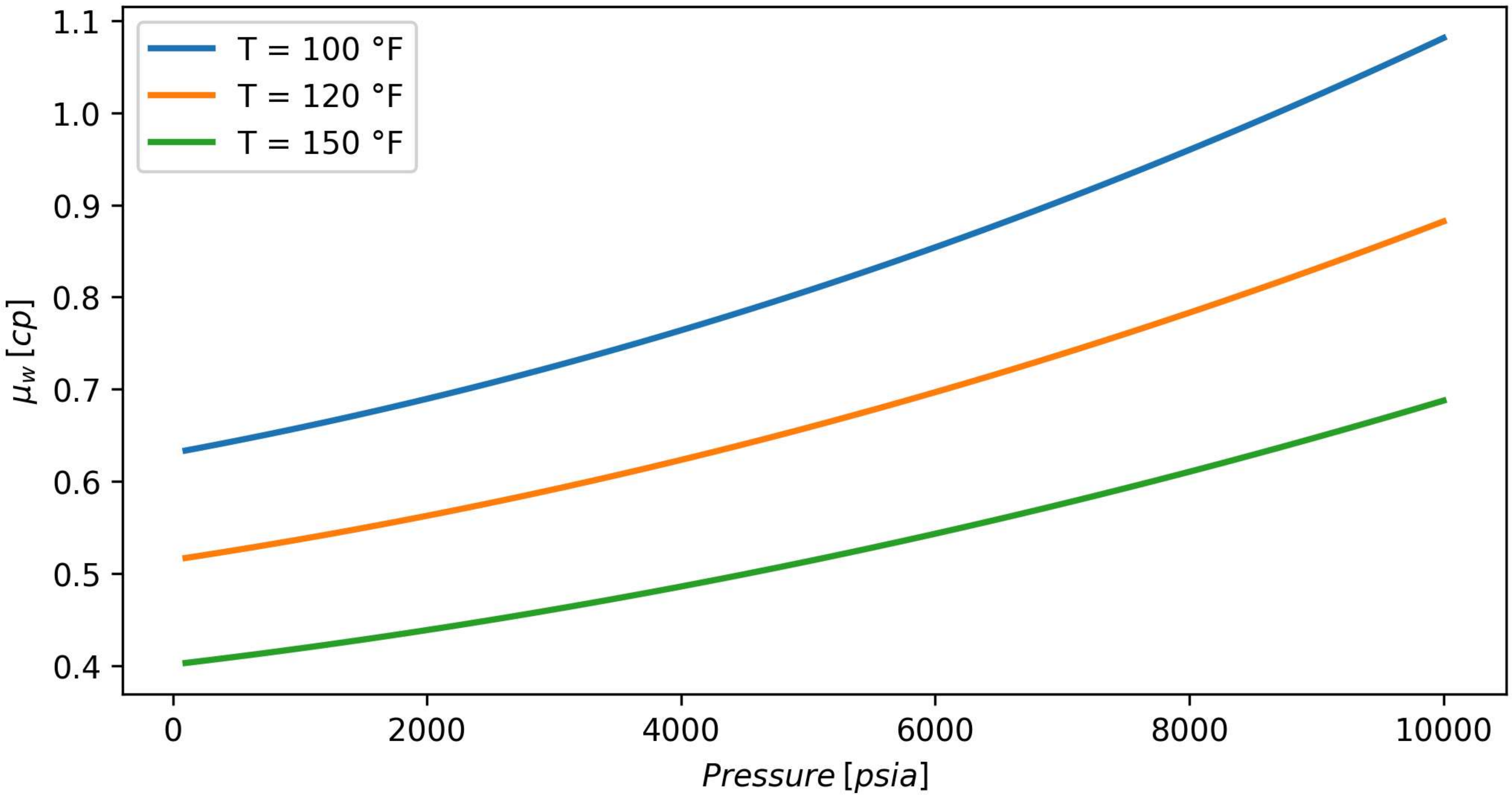
In [25]:
```
import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

In [26]:
```
T=np.array([100,120,150])
S=0.2
p=np.linspace(100,10000,100)
```

2.3 Calculation and plotting

In [29]:
```
plt.figure(dpi=300, figsize=(8,4))


for i in range (len(T)):

    A=109.574-8.40564*S+0.313314*S**2+8.72213*10**-3*S**3
    B=-1.12166+2.63951*10**-2*S-6.79461*10**-4*S**2-5.47119*10**-5*S**3+1.55586*10**-6*S**4
    muw1=A*T[i]**B

    muw_values=[]

    for j in range (len(p)):

        muw=(0.9994+4.0295*10**-5*p[j]+3.1062*10**-9*p[j]**2)*muw1
        muw_values.append(muw)



    plt.plot(p, muw_values, linestyle='-', linewidth=2,
             label=f'T = {T[i]} °F')

plt.xlabel('$Pressure\:[psia]$')
plt.ylabel('$\mu_w\:[cp]$')
# plt.grid(True)
plt.legend()
plt.show()
```



3. Conclusions and highlights

```
In [ ]:
```

4. Comments

I hope this notebook is useful in your journey of coding appplied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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In [ ]:
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